

Faculty of Computing



[Computer Communications & Network]

Lab No 4 Tasks

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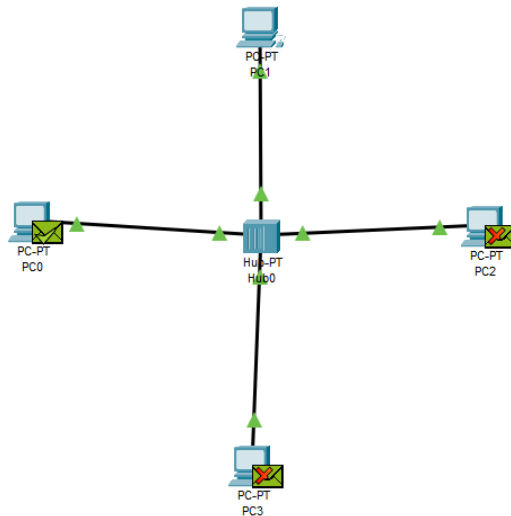
SAP 64281

Task 1: Connect 4 PCs with the HUB in cisco packet tracer and send data from PC 1 to PC 2 and check the working of HUB through simulation mode. (05 Marks)

Steps of Task 1: Connect 4 PCs with a HUB in Cisco Packet Tracer

1. **Open Cisco Packet Tracer:** Launch the application.
2. **Add Devices:**
 - Click on the "**End Devices**" icon (the one that looks like a PC) at the bottom left.
 - Place **four PCs** in the workspace. You can do this by clicking the PC icon and then clicking in the main area four times.
 - Click on the "**Network Devices**" icon (the one with the router and switch).
 - Select the "**Hub**" icon and place a hub in the center of your PCs.
3. **Connect Devices:**
 - Click on the "**Connections**" icon (the lightning bolt).
 - Select the "**Copper Straight-Through**" cable (the solid black line).
 - Click on the first PC, select **FastEthernet0**, then click on the hub and select one of the available ports (e.g., FastEthernet0/1). Repeat this process to connect all four PCs to the hub.
4. **Configure IP Addresses:**
 - Click on **PC1**, go to the "**Desktop**" tab, and then select "**IP Configuration.**"
 - Enter a unique IP address for each PC, such as 192.168.1.1 for PC1, 192.168.1.2 for PC2, and so on. Use the same subnet mask (255.255.255.0) for all of them.
5. **Send Data and Observe (Simulation Mode):**
 - Switch from "**Realtime**" to "**Simulation**" mode in the bottom right corner of the window.
 - Click on the "**Add Simple PDU**" tool (the envelope icon) on the right side of the toolbar.
 - Click on **PC1** as the source and then on **PC2** as the destination.
 - Click the "**Capture/Forward**" button (the "play" button with an arrow) to watch the simulation.

What you will see: The packet will travel from PC1 to the hub. The hub, being a layer-1 device, does not know the destination and simply broadcasts the packet to **all other connected PCs** (PC2, PC3, and PC4). Only PC2 will accept the packet, and the others will discard it.



Task 2: Connect 4 PCs with the SWITCH in cisco packet tracer and send data from PC 1 to PC 2 and check the working of SWITCH through simulation mode. (05 Marks)

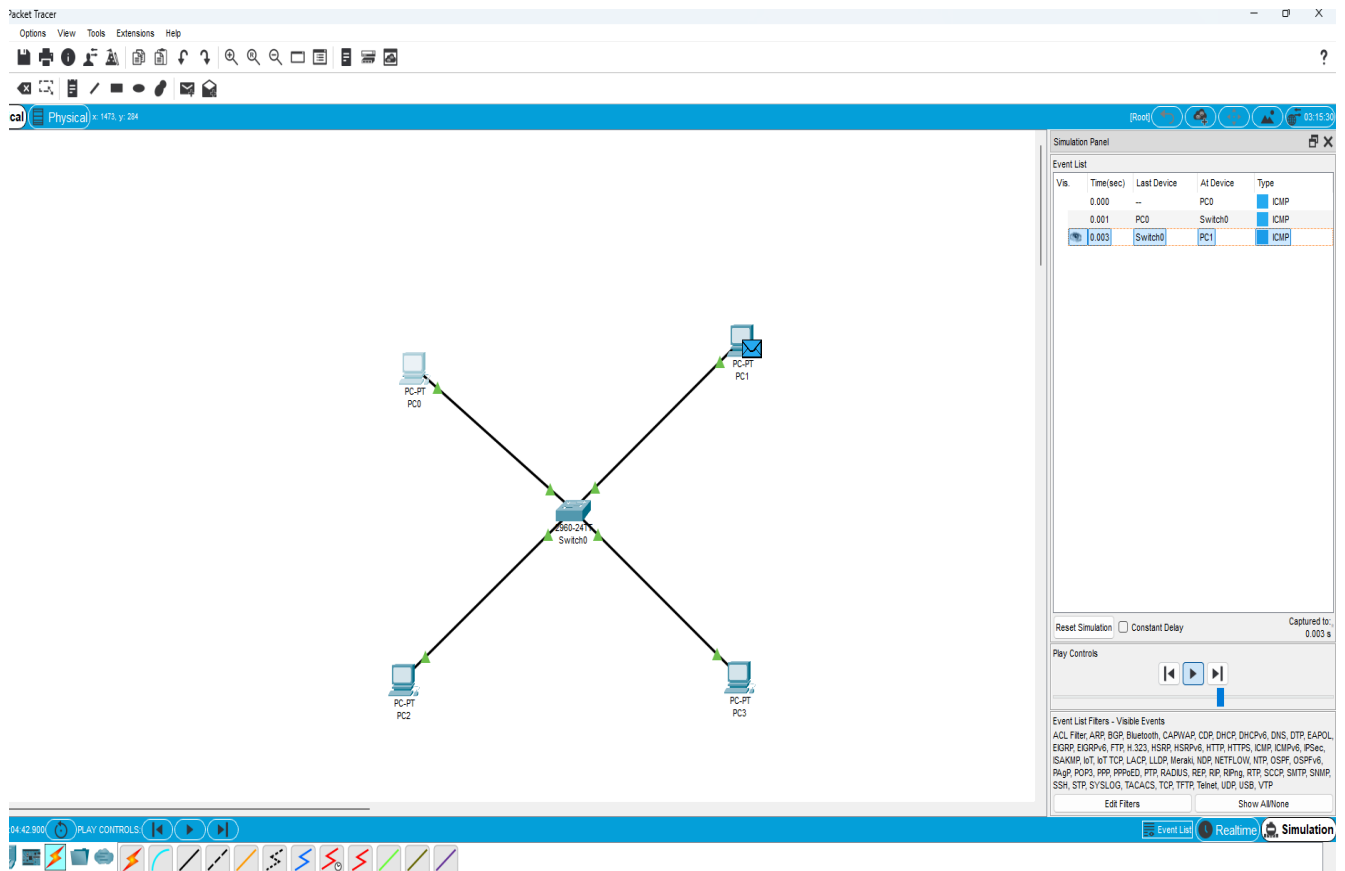
Task 2: Connect 4 PCs with a SWITCH in Cisco Packet Tracer

1. **Open Cisco Packet Tracer:** Launch the application.
2. **Add Devices:**
 - Add **four PCs** to the workspace.
 - Click on the "**Network Devices**" icon and select the "**Switch**" icon. Place a switch in the center.
3. **Connect Devices:**
 - Click on the "**Connections**" icon.
 - Select the "**Copper Straight-Through**" cable.
 - Connect each PC to an available port on the switch (e.g., connect PC1 to the switch's FastEthernet0/1, PC2 to FastEthernet0/2, and so on).
4. **Configure IP Addresses:**
 - Just like in Task 1, configure a unique IP address and the same subnet mask (192.168.1.1 to 192.168.1.4, with subnet mask 255.255.255.0) for each PC.
5. **Send Data and Observe (Simulation Mode):**
 - Switch to "**Simulation**" mode.

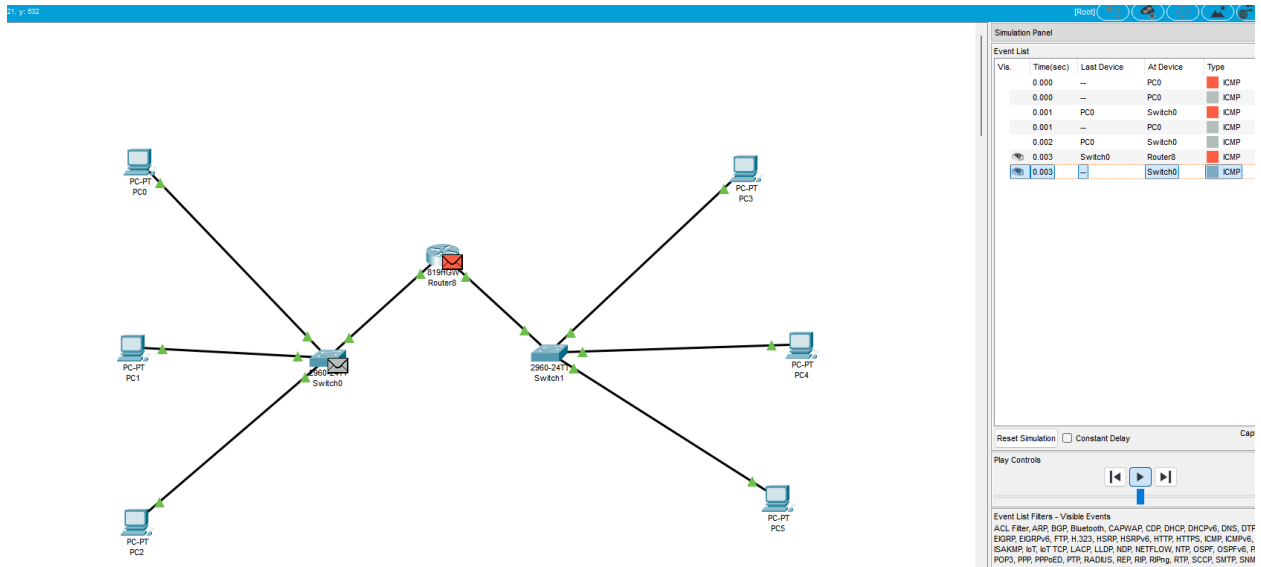
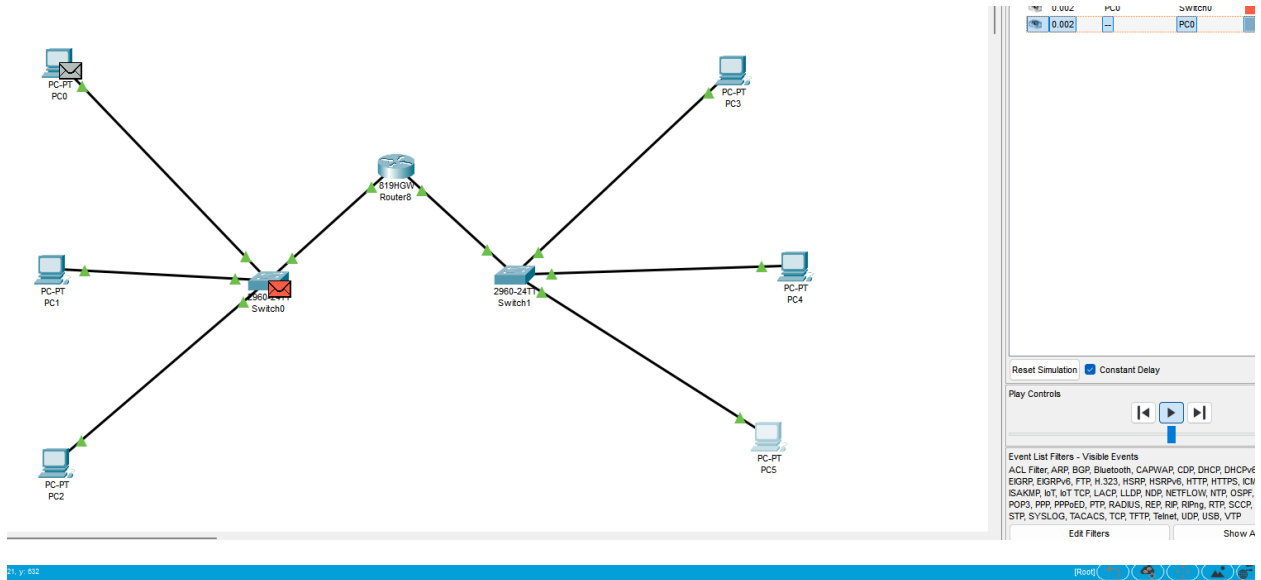
- Click the "**Add Simple PDU**" tool.
- Click on **PC1** as the source and then on **PC2** as the destination.
- Click the "**Capture/Forward**" button.

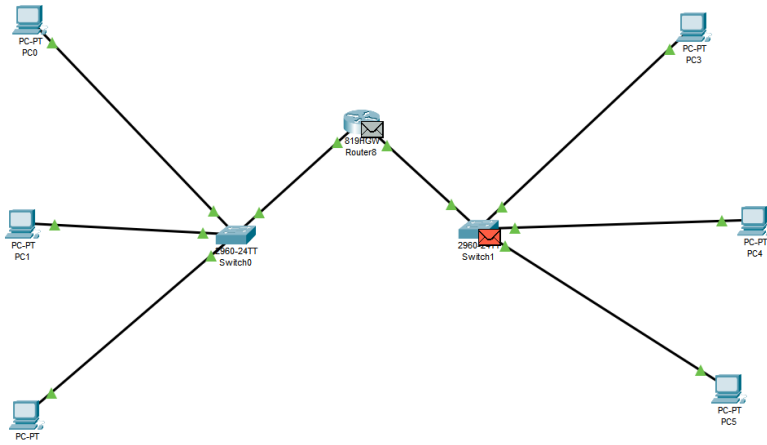
What you will see: The packet will travel from PC1 to the switch. The first time you send a packet, the switch, as a layer-2 device, will broadcast it to all other connected devices just like a hub. This is because it doesn't know the destination's MAC address yet. However, once PC2 replies, the switch will learn its MAC address and associate it with the port it's connected to. Subsequent packets from PC1 to PC2 will be sent **directly to PC2 only**, as the switch now knows the specific port for that destination. The other PCs will not receive the packet. This is the key difference between a hub (a broadcast device) and a switch (a more intelligent device).

Note: Save all the cisco packets files for reference in the next lab. Create a word document and attach screenshots of all the simulations under each task for evaluation.



TASK 03:



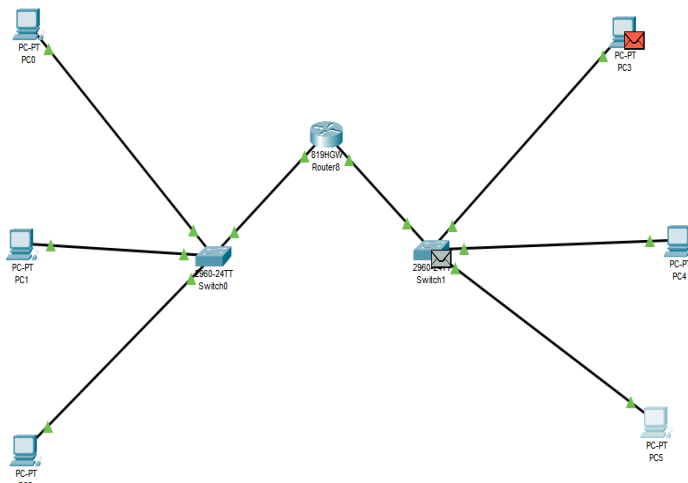


Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.001	--	PC0	ICMP
	0.002	PC0	Switch0	ICMP
	0.002	Switch0	Router8	ICMP
	0.003	Switch0	Router8	ICMP
	0.003	Router8	Switch1	ICMP

Reset Simulation ☒ Constant Delay Ca

Play Controls

Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, D, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6

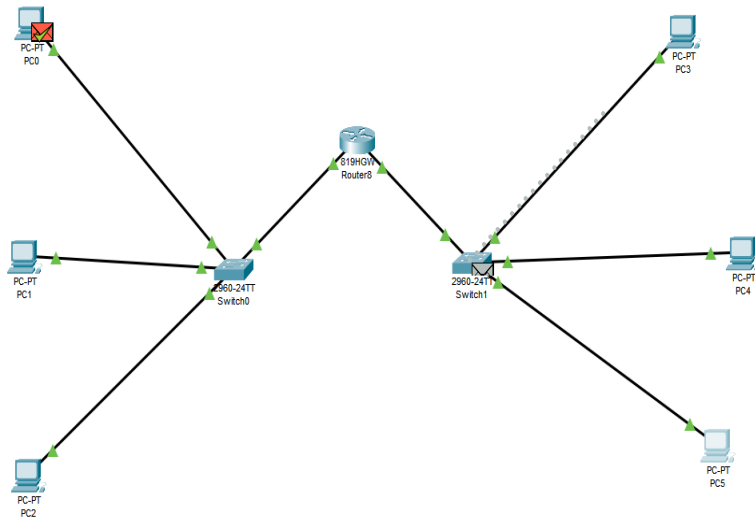


Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.000	--	PC0	ICMP
	0.001	PC0	Switch0	ICMP
	0.001	--	PC0	ICMP
	0.002	PC0	Switch0	ICMP
	0.002	Switch0	Router8	ICMP
	0.003	Switch0	Router8	ICMP
	0.003	Router8	Switch1	ICMP
	0.004	Router8	Switch1	ICMP
	0.004	Switch1	PC3	ICMP

Reset Simulation ☒ Constant Delay Captured to: 0.004 s

Play Controls

Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IRT, BT, TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPv2, RIPv3, SCCP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP



THE END

0.000	--	PC0	ICMP
0.001	PC0	Switch0	ICMP
0.001	--	PC0	ICMP
0.002	PC0	Switch0	ICMP
0.003	Switch0	Router8	ICMP
0.003	--	Switch0	ICMP
0.004	Router8	Switch1	ICMP
0.005	Switch0	Router8	ICMP
0.006	Switch1	PC3	ICMP
0.007	PC3	Switch1	ICMP
0.008	Router8	Switch1	ICMP
0.009	Switch1	Router8	ICMP
0.010	Switch1	PC3	ICMP
0.010	Router8	Switch0	ICMP
0.012	Switch0	PC0	ICMP

☐ Constant Delay
 Capt

Play Controls

Event List Filters - Visible Events
 ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, ISAKMP, IoT TCP, LACP, LLDP, NDP, NETFLOW, NTP, OSPF, OSPFv6, P, POP3, PPP, PPPoE, PTP, RADIUS, REP, RP, RRPng, RTP, SCCP, SMTP, SNMP